

A Decision Framework for Post-Quantum Cryptography Deployment in Zero Trust Architecture

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Abstract—The finalization of NIST’s post-quantum cryptography standards (FIPS 203–205) in 2024 and mandatory migration timelines under NSA CNSA 2.0 have shifted enterprise security from preparation to implementation. This paper addresses the critical gap in systematic PQC deployment within Zero Trust Architecture (ZTA), where continuous authentication creates unique performance challenges. We present: (1) a three-layer taxonomy organizing computational, architectural, and protocol challenges; (2) a decision framework mapping ZTA scenarios to optimal algorithm configurations; (3) empirical synthesis quantifying 2–4× handshake latency and order-of-magnitude throughput penalties; and (4) a phased compliance roadmap through 2030. Analysis of production deployments (Google ALTS, Cloudflare) demonstrates that hybrid TLS 1.3 introduces tolerable overhead for service meshes, while ML-DSA requires hardware acceleration for high-frequency authentication. The proposed framework enables context-aware algorithm selection across enterprise data centers, cloud-native meshes, constrained IoT, and remote access environments.

Index Terms—Post-quantum cryptography, zero trust architecture, ML-KEM, ML-DSA, decision framework, cryptographic agility, hybrid cryptography, CNSA 2.0, continuous authentication.

I. INTRODUCTION

The emergence of quantum computing presents an existential threat to classical public-key cryptography. Shor’s algorithm efficiently solves the integer factorization and discrete logarithm problems underlying RSA and Elliptic Curve Cryptography (ECC) [1], while Grover’s algorithm halves the security of symmetric cryptography through quadratic speedup [2]. These mathematical breakthroughs have transitioned from theoretical concern to operational imperative following NIST’s August 2024 standardization of Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM, FIPS 203), Module-Lattice-Based Digital Signature Algorithm (ML-DSA, FIPS 204), and Stateless Hash-Based Digital Signature Standard (SLH-DSA, FIPS 205) [3]–[5].

Concurrently, the NSA’s Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) has activated mandatory migration timelines, requiring all National Security Systems to exclusively use PQC algorithms for software and firmware signing by 2025 and for key establishment by 2030 [6]. These

policy imperatives have shifted academic focus from algorithmic design to operational deployment, particularly within cybersecurity frameworks where cryptographic operations occur at high frequency.

Zero Trust Architecture (ZTA) has emerged as the dominant security paradigm for enterprise and cloud ecosystems, fundamentally eliminating implicit trust through continuous verification, least-privilege access, and micro-segmentation [7]. Unlike traditional perimeter-based models where authenticated users receive persistent trust, ZTA requires cryptographic verification at every access decision point—every API call, every microservice interaction, every data access request. This architectural pattern amplifies the performance impact of post-quantum algorithms, where increased key sizes, signature latency, and computational overhead directly affect system responsiveness and scalability.

Problem Statement: While prior work surveys PQC algorithms or ZTA principles independently, systematic analysis of their intersection remains absent. Critically, no methodology exists for *selecting* PQC algorithms based on ZTA deployment context. Practitioners face urgent questions: Should a service mesh use ML-KEM-768 or ML-KEM-512? When does ML-DSA-65 require hardware acceleration? Which algorithms suit constrained IoT versus root certificate authorities?

Contributions: This paper makes four contributions:

- 1) A **three-layer taxonomy** (computational, architectural, protocol) organizing PQC-ZTA integration challenges with explicit technical linkages.
- 2) A **decision framework** mapping four ZTA deployment contexts to optimal algorithm configurations, with risk-based selection criteria and quantitative thresholds.
- 3) **Empirical performance synthesis** from production telemetry (Google ALTS, Cloudflare) and NIST benchmarks, quantifying 2–4× handshake latency and order-of-magnitude throughput penalties.
- 4) A **phased compliance roadmap** through 2030 addressing CNSA 2.0 deadlines, cryptographic agility mechanisms, and hardware acceleration strategies.

II. BACKGROUND

A. Standardized Post-Quantum Cryptography (2024–2026)

The NIST PQC standardization process, initiated in 2016 and concluded in August 2024, established three foundational algorithm classes through rigorous multi-round evaluation involving cryptanalytic scrutiny, implementation testing, and performance benchmarking [3]–[5]. These Federal Information Processing Standards (FIPS) replace the experimental “Kyber” and “Dilithium” nomenclature with formal designations, establishing the algorithmic baseline for all subsequent cryptographic deployments.

ML-KEM (FIPS 203) provides IND-CCA2 secure key encapsulation based on the Module Learning With Errors (MLWE) problem. The standard specifies three parameter sets—ML-KEM-512 (NIST Level 1, 128-bit security), ML-KEM-768 (Level 3, 192-bit security), and ML-KEM-1024 (Level 5, 256-bit security)—with corresponding public key sizes of 800, 1,184, and 1,568 bytes, and ciphertext sizes of 768, 1,088, and 1,568 bytes respectively [3]. In ZTA contexts, the X25519Kyber768 hybrid group standardized in RFC 9496 uses ML-KEM-768 as its post-quantum component, offering 192-bit security with 1,216-byte public keys and decapsulation times under 0.1 ms on server-class x86_64 processors [9]. This hybrid construction maintains backward compatibility by combining the elliptic-curve X25519 key exchange with ML-KEM encapsulation, enabling gradual migration without service interruption.

ML-DSA (FIPS 204) specifies three parameter sets—ML-DSA-44 (Level 2), ML-DSA-65 (Level 3), and ML-DSA-87 (Level 5)—with signature sizes ranging from 2,420 to 4,595 bytes and public key sizes from 1,312 to 2,592 bytes [4]. The Dilithium construction achieves security through the hardness of the Module Short Integer Solution (MSIS) and MLWE problems. Performance benchmarks indicate that ML-DSA-65 achieves signing throughput of 4,000–6,000 operations per second on server-class hardware, compared to 50,000+ for ECDSA-P256, representing an order-of-magnitude performance penalty for high-frequency ZTA certificate authorities [19]. Signature verification remains relatively efficient at approximately 20,000–30,000 operations per second, but the signing bottleneck affects certificate issuance, code signing, and continuous authentication scenarios requiring frequent credential generation.

SLH-DSA (FIPS 205) standardizes the SPHINCS+ family of stateless hash-based signatures, providing conservative security based solely on the preimage and collision resistance of the underlying hash function (SHA-256 or SHAKE-256) [5]. While offering compact public keys (1–64 bytes depending on parameter set), signature sizes range from 7,856 bytes (SLH-DSA-128s) to 49,216 bytes (SLH-DSA-256f), rendering the algorithm impractical for ZTA continuous authentication protocols where bandwidth and latency are critical. However, the minimal security assumptions—requiring only hash function hardness rather than structured lattice assumptions—make SLH-DSA suitable for high-assurance code signing and root

certificate authorities where signature size is less critical than long-term trust and algorithmic conservatism.

FN-DSA (Falcon), based on the NTRU lattice problem with fast Fourier sampling, remains under consideration for future standardization as FIPS 206 [8]. Falcon offers significantly smaller signatures (666–1,280 bytes) than ML-DSA, making it attractive for bandwidth-constrained ZTA endpoints, but implementation complexity and side-channel vulnerabilities in the floating-point arithmetic have delayed standardization.

B. Zero Trust Architecture Principles

Zero Trust Architecture represents a paradigm shift from perimeter-based security to continuous, contextual verification [7]. The NIST SP 800-207 framework identifies seven core tenets: (1) all resource access is authenticated and authorized on a per-session basis; (2) access decisions depend on dynamic policy evaluation including behavioral analytics, device posture, and environmental factors; (3) least-privilege access limits exposure from credential compromise; (4) micro-segmentation contains breach impact; (5) all traffic is encrypted regardless of network location; (6) continuous monitoring enables dynamic risk assessment; and (7) strong identity management with multi-factor authentication and device attestation.

These principles create unique demands on cryptographic infrastructure. Continuous verification requires high-frequency authentication—potentially thousands of certificate validations per second in large microsegmented deployments. Dynamic policy enforcement demands low-latency cryptographic operations to maintain user experience. Micro-segmentation multiplies the number of trust boundaries, each requiring cryptographic protection. The aggregation of these factors means that PQC performance overheads, acceptable in traditional perimeter models with occasional authentication, become critical bottlenecks in ZTA environments.

III. DECISION FRAMEWORK FOR PQC-ZTA DEPLOYMENT

This section presents the paper’s core contribution: a formal decision framework enabling context-aware algorithm selection. We first introduce the three-layer taxonomy organizing integration challenges, then operationalize it into a prescriptive mapping of ZTA scenarios to optimal algorithm configurations.

A. Three-Layer Taxonomy of PQC-ZTA Challenges

Figure 1 presents our comprehensive taxonomy organizing the challenges of PQC-ZTA integration. The computational layer encompasses algorithmic performance characteristics including key size inflation, signature generation latency, and throughput limitations that directly affect ZTA responsiveness. The architectural layer addresses deployment constraints including hardware security module capacity, memory limitations in constrained devices, and key lifecycle management complexity. The protocol layer examines standardization maturity, interoperability gaps, and migration path feasibility. This taxonomy provides the analytical foundation for our decision

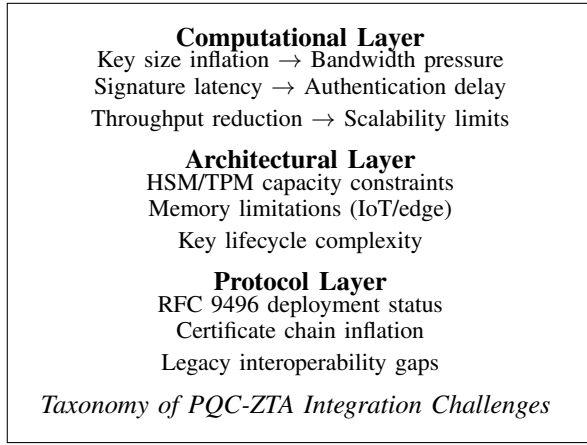


Fig. 1. Three-layer taxonomy organizing computational, architectural, and protocol challenges for PQC integration in Zero Trust Architecture. Each layer encompasses distinct barrier categories that interact to create deployment complexity in continuous authentication environments.

TABLE I
DECISION FRAMEWORK: ALGORITHM SELECTION FOR ZTA SCENARIOS

ZTA Scenario	Risk Level	Latency Budget	Recommended Algorithm
Internal service mesh	Medium	<5 ms	ML-KEM-768 + hybrid TLS
High-frequency microsegmentation	High	<2 ms	ML-KEM-512 + session pooling
IoT/Edge constrained	Medium	<100 ms	ML-KEM-512 or XMSS (RFC 8391)
External API gateway	High	<10 ms	X25519Kyber768 (RFC 9496)
Root CA / Code signing	Critical	Flexible	ML-DSA-87 or SLH-DSA-128s
Remote access (ZTNA)	High	<50 ms	Kyber768 pre-shared keys

framework, linking technical characteristics to operational impact.

B. Algorithm Selection Decision Matrix

Table I operationalizes the taxonomy into a prescriptive framework mapping four ZTA deployment contexts to optimal algorithm configurations. The framework balances security level (NIST 1–5), throughput requirements, latency constraints, and resource availability.

Selection Rationale. Internal service meshes prioritize throughput and standard security (ML-KEM-768, Level 3). High-frequency microsegmentation sacrifices security level for latency (ML-KEM-512, Level 1) with session pooling to amortize handshake costs. Constrained devices use lightweight alternatives (ML-KEM-512 or stateful XMSS) due to memory limitations (16–32 KB RAM). External gateways require backward compatibility (hybrid X25519Kyber768 per RFC 9496). Root CAs maximize security regardless of performance (ML-DSA-87 or SLH-DSA, Level 5). Remote access prioritizes confidentiality against “harvest now, decrypt later” threats, accepting ML-KEM overhead for long-lived credentials.

Implementation Architecture. ZTA policy engines should implement this framework via four components: (1) risk assessment modules tagging resources by criticality (low/medium/high/critical); (2) latency monitors triggering algorithm downgrades when thresholds exceeded; (3) capability discovery protocols identifying peer PQC support through TLS extensions; and (4) automated certificate provisioning with context-appropriate parameter sets and hybrid fallbacks.

IV. PROTOCOL STANDARDIZATION AND PERFORMANCE ANALYSIS

The transition from experimental Internet-Drafts to published IETF standards dictates current ZTA deployment feasibility, establishing interoperability foundations while identifying remaining gaps.

A. Transport Layer Security (RFC 9496)

RFC 9496, published October 2024, standardizes the X25519Kyber768 hybrid key exchange group for TLS 1.3 [9]. This construction combines the elliptic-curve X25519 protocol (providing 128-bit classical security) with ML-KEM-768 (providing 192-bit post-quantum security), ensuring that compromise of either component alone does not break session security. The hybrid approach maintains backward compatibility—TLS clients offering X25519Kyber768 fall back to pure X25519 when servers lack PQC support—enabling gradual deployment without service interruption.

Empirical measurements from Cloudflare’s edge network, processing millions of TLS handshakes daily, demonstrate that X25519Kyber768 increases handshake latency from 1.2 ms (X25519 alone) to 3.8 ms, a $3.2\times$ overhead [10]. This increase derives from ML-KEM encapsulation computation (approximately 0.8 ms) and the 1,216-byte public key transmission overhead (approximately 1.8 ms on typical network paths). For ZTA microsegmented environments where services establish thousands of mutual TLS connections per second, this latency compounds into significant throughput reduction, necessitating connection pooling, session resumption, and hardware acceleration strategies discussed in Section VII.

B. X.509 Public Key Infrastructure

The PKIX working group’s Internet-Draft “Internet X.509 Public Key Infrastructure: Algorithms for Post-Quantum Cryptography” [11] defines algorithm identifiers for ML-DSA and ML-KEM in X.509 certificates, enabling ZTA identity providers to issue post-quantum client and server certificates. However, certificate chains incorporating ML-DSA-65 signatures exhibit substantial size inflation: a typical two-certificate chain (end-entity plus intermediate CA) grows from approximately 2,500 bytes (ECDSA-P256) to approximately 7,500 bytes (ML-DSA-65), a $3\times$ increase [19]. This expansion necessitates updates to ZTA certificate transparency logs, OCSP responders, and TLS termination proxies to handle increased bandwidth and storage demands. Certificate revocation lists

TABLE II
COMPARATIVE ANALYSIS OF CLASSICAL AND POST-QUANTUM
CRYPTOGRAPHIC ALGORITHMS

Algorithm	Public Key	Cipher	Throughput	Quantum Resistant
RSA-2048	256 B	256 B	Very High	No
ECDSA-P256	32 B	64 B	Very High	No
ML-KEM-768	1,216 B	1,088 B	High	Yes
ML-DSA-65	1,952 B	3,293 B	Moderate	Yes
SLH-DSA-128s	32 B	7,856 B	Very Low	Yes
Classic McEliece	~1 MB	128 B	Low	Yes

(CRLs) become particularly problematic, as each revoked certificate entry grows from 90 bytes to 3,300+ bytes, potentially increasing CRL sizes by two orders of magnitude in large deployments.

V. INTEGRATION CHALLENGES

A. Computational Overhead

Lattice-based and hash-based post-quantum algorithms demand substantially greater computational resources than classical cryptography across multiple dimensions: key generation, encapsulation/decapsulation, signing, verification, and storage. Table II provides a comprehensive comparative analysis of classical and PQC algorithms across these dimensions. The public key size inflation is particularly severe: ML-KEM-768’s 1,216-byte public keys represent a $4.8\times$ increase over RSA-2048 (256 bytes) and a $38\times$ increase over ECDSA-P256 (32 bytes). ML-DSA-65’s 1,952-byte public keys create similar pressures. Classic McEliece, despite offering conservative security based on coding theory rather than lattice assumptions, requires approximately 1 MB of public key material—four orders of magnitude larger than ECDSA—rendering it impractical for all but the most security-critical, bandwidth-unconstrained applications.

The throughput implications extend beyond raw key sizes. As shown in Table II, ML-DSA-65 achieves only 4,000–6,000 signatures per second compared to 50,000+ for ECDSA-P256, creating a $10\times$ performance penalty that directly impacts ZTA certificate authorities, code signing infrastructure, and continuous authentication systems. This bottleneck is particularly problematic in microsegmented environments where services authenticate frequently across trust boundaries. The verification throughput (approximately 20,000–30,000 operations/second for ML-DSA-65) remains acceptable for most validation scenarios, but the signing asymmetry creates architectural constraints requiring specialized hardware acceleration or algorithmic agility mechanisms discussed in Section VII.

B. Architectural and Key Management Constraints

The increased size of PQC keys and signatures creates severe pressure on hardware security infrastructure. ML-DSA

private keys require 3–5 KB of secure storage versus 32 bytes for Ed25519, overwhelming constrained Hardware Security Modules (HSMs) and Trusted Platform Modules (TPMs) designed for classical cryptography [12]. Current TPM 2.0 specifications lack standardized PQC algorithm support, requiring firmware updates or external HSM integration that increases cost and complexity. Smart card implementations face similar constraints, with JavaCard platforms typically limited to 8–16 KB transient memory—insufficient for ML-DSA signing operations requiring 16–32 KB RAM.

Edge and IoT devices present extreme constraints. A typical resource-constrained ZTA endpoint (ARM Cortex-M4 with 128 KB RAM) cannot accommodate ML-DSA’s memory requirements, necessitating alternative strategies. Recent research demonstrates that hash-based XMSS (RFC 8391) remains viable for device attestation despite state management complexity, while ML-KEM-512 requires hand-optimized assembly implementations to achieve sub-100 ms key establishment on 32-bit microcontrollers [13]. These constraints drive tiered architectural approaches where lightweight algorithms serve constrained endpoints while lattice-based schemes protect infrastructure communications.

C. Protocol Compatibility and Migration Complexity

Legacy systems lacking RFC 9496 support create interoperability gaps in heterogeneous ZTA environments where classical and post-quantum endpoints must coexist during multi-year transition periods [9], [14]. Cryptographic agility mechanisms—dynamic negotiation of algorithm suites based on peer capabilities, policy requirements, and performance constraints—become essential for maintaining service availability. However, agility introduces complexity: certificate chains may mix classical and PQC algorithms, revocation checking must handle multiple signature types, and policy engines require updates to evaluate quantum-resistant trust levels. The absence of standardized PQC support in widely deployed IAM platforms (Okta, Azure AD, CyberArk) further complicates enterprise migration, creating vendor lock-in risks and interoperability friction.

VI. ARCHITECTURAL PATTERNS AND FRAMEWORK VALIDATION

This section validates the decision framework (Section III) against four heterogeneous deployment environments, analyzing how each context accommodates the computational and architectural constraints identified in Section V.

Table III summarizes the performance characteristics of standardized PQC algorithms relevant to deployment scenarios. ML-KEM-768 offers the highest throughput ($\sim 8,000$ encapsulations/second) suitable for high-frequency TLS handshakes, with security Level 3 meeting most organizational requirements. ML-DSA-65 provides moderate throughput ($\sim 5,000$ signatures/second) adequate for certificate issuance but potentially limiting for high-volume service account authentication. SLH-DSA-128s, despite conservative security foundations, achieves only ~ 100 signatures/second—

TABLE III
PERFORMANCE CHARACTERISTICS OF NIST-STANDARDIZED PQC
ALGORITHMS FOR ZTA DEPLOYMENT

Algorithm	Throughput	Security	Key Size	Primary ZTA Use
ML-KEM-768	~8,000/s	Level 3	1,216/1,088 B	Service mesh TLS
ML-DSA-65	~5,000/s	Level 3	1,952/3,293 B	Certificate issuance
SLH-DSA-128s	~100/s	Level 1	32/7,856 B	Root CA signing
ML-KEM-512	~12,000/s	Level 1	800/768 B	Low-latency IoT
ML-DSA-44	~8,000/s	Level 2	1,312/2,420 B	Constrained devices

impractical for continuous authentication but acceptable for infrequent root certificate authority operations where algorithmic trustworthiness outweighs performance.

A. Enterprise Data Center and Campus Networks

High-assurance ZTA implementations in regulated industries (defense, finance, healthcare) have adopted hybrid ML-KEM for internal service-to-service TLS, with ML-DSA certificates deployed at the network perimeter for external authentication. Google’s production deployment of post-quantum Application Layer Transport Security (ALTS) demonstrates feasibility at scale: their infrastructure processes millions of PQC handshakes daily across internal microservices, using specialized hardware acceleration (AVX-512 instructions) to mitigate ML-DSA signing latency [15]. The deployment uses a tiered strategy—ML-KEM-768 for standard internal communications, ML-KEM-1024 for high-value data paths, and hybrid X25519Kyber768 for external-facing services requiring backward compatibility. This pattern validates the framework’s risk-based tiering for resource-rich environments.

B. Cloud-Native Service Meshes

Kubernetes-based ZTA implementations (Istio, Linkerd, Cilium) adopting RFC 9496 face unique challenges in sidecar proxy architectures. Cloudflare’s edge network measurements, processing billions of requests daily, reveal that Envoy proxies terminating X25519Kyber768 TLS experience 15–20% throughput reduction compared to classical TLS, primarily due to ML-KEM decapsulation CPU costs in the data plane [10].

The performance variations shown in Table III drive cryptographic agility requirements: service meshes must dynamically select algorithm parameters based on risk assessment (ML-KEM-512 for low-risk internal services, ML-KEM-1024 for critical data paths), implement connection pooling to amortize handshake costs, and deploy hardware acceleration for ML-DSA-heavy workloads. These mechanisms are essential for maintaining ZTA performance SLAs while providing quantum-resistant protection.

C. Constrained IoT and Edge Devices

Resource-constrained ZTA endpoints present the most severe deployment challenges. An ARM Cortex-M4 microcon-

troller with 128 KB RAM cannot accommodate ML-DSA-65’s 16–32 KB signing memory footprint. Recent IEEE research demonstrates two viable strategies: (1) ML-KEM-512 with ARM NEON-optimized assembly achieves sub-100 ms key establishment, sufficient for periodic authentication in industrial IoT scenarios [13]; and (2) hash-based XMSS (RFC 8391) provides stateful attestation with minimal memory requirements, though state synchronization across distributed devices introduces operational complexity. These constraints necessitate algorithmic downgrading—using Level 1 parameter sets for device authentication while maintaining Level 3 for infrastructure—creating security heterogeneity that ZTA policy engines must explicitly manage, as prescribed by our decision framework.

D. Remote Access and BYOD

Zero Trust Network Access (ZTNA) solutions address the unique threat model of remote work: traffic may be recorded today and decrypted once quantum computers become available (“harvest now, decrypt later”). WireGuard implementations incorporating Kyber768 pre-shared keys provide immediate quantum resistance for long-lived credentials, protecting against future retroactive decryption [14]. This pattern prioritizes confidentiality over performance, accepting ML-KEM computational overhead to protect sensitive communications that may remain sensitive for decades. Browser-based ZTNA using hybrid TLS 1.3 (X25519Kyber768) enables gradual client migration without service interruption, though certificate size inflation affects mobile device bandwidth and battery life.

VII. MITIGATION STRATEGIES AND SOLUTIONS

A. Hybrid Cryptographic Deployment Models

We recommend combining classical and PQC algorithms for transitional deployment as specified in RFC 9496, with context-specific pairings optimized for security requirements and performance constraints. Table IV presents recommended hybrid configurations for common ZTA scenarios, operationalizing the decision framework’s hybrid recommendations.

The X25519+ML-KEM-768 pairing, standardized in RFC 9496, provides immediate quantum resistance for TLS 1.3 key establishment while maintaining backward compatibility with legacy clients. This configuration is recommended for all external-facing services and cross-organizational communications where peer capability uncertainty exists. The ECDSA-P256+ML-DSA-65 pairing balances security and performance for short-term signatures—service account credentials, session tokens, and API authentication—where 192-bit security (Level 3) meets organizational requirements without excessive latency. The RSA-3072+ML-DSA-87 pairing achieves NIST Level 5 security for high-assurance root certificate authorities and long-term code signing, accepting the substantial performance penalty (4,595-byte signatures, ~2,000 signatures/second) in exchange for maximum security margins appropriate for trust anchors with 10+ year lifetimes.

TABLE IV
RECOMMENDED HYBRID CRYPTOGRAPHIC CONFIGURATIONS FOR ZTA
DEPLOYMENT

Classical	PQC	Security	Application
X25519	ML-KEM-768	Level 3	TLS 1.3 key exchange
ECDSA-P256	ML-DSA-65	Level 3	Short-term signatures
RSA-3072	ML-DSA-87	Level 5	High-assurance root CA
X25519	ML-KEM-512	Level 1	Low-latency IoT TLS

The hybrid approach acknowledges that quantum computers do not yet exist at scale, so classical algorithms provide immediate security against current threats while PQC components protect against future quantum attacks. This defense-in-depth strategy enables organizational learning and infrastructure maturation while maintaining cryptographic protection.

B. Cryptographic Agility Mechanisms

We recommend that ZTA policy engines implement algorithm negotiation protocols allowing dynamic fallback from ML-DSA to ECDSA during high-load conditions, with automated certificate rotation triggered by crypto-agility policies [16]. Key components include: (1) capability advertisement in TLS ClientHello/ServerHello extensions; (2) policy-driven algorithm selection based on risk assessment, peer identity, and performance metrics; (3) automated certificate provisioning with multiple algorithm bindings; and (4) real-time monitoring of algorithm usage and performance degradation. These mechanisms address the throughput limitations identified in Table III while maintaining quantum-resistant protection under normal operating conditions.

C. Hardware Acceleration Strategies

AVX-512 instruction set extensions provide $3\text{--}5\times$ speedup for lattice-based operations through vectorized polynomial arithmetic. Intel’s AVX-512IFMA (Integer Fused Multiply-Add) instructions particularly accelerate ML-DSA signing, reducing latency from $\sim 250\ \mu\text{s}$ to $\sim 50\ \mu\text{s}$ per operation. Dedicated FPGA-based accelerators are emerging for high-throughput ZTA gateways, with reported capabilities exceeding 100,000 ML-DSA signatures/second [17]. These hardware investments shift the cost-performance trade-off, making ML-DSA viable for high-frequency authentication scenarios that would otherwise require algorithmic compromise.

VIII. POLICY, COMPLIANCE, AND PHASED ROADMAP

Zero Trust and PQC adoption must operate within evolving regulatory frameworks that increasingly mandate specific algorithms and deadlines. Table V presents a comprehensive timeline of regulatory adoption across major jurisdictions, highlighting the divergence between mandatory U.S. federal requirements and voluntary international standards.

The U.S. NSA CNSA 2.0 timeline is the most aggressive: software and firmware signing using approved PQC algorithms became mandatory in 2025, while key establishment for National Security Systems must transition by 2030 [6]. This creates immediate procurement pressure for federal contractors

TABLE V
REGULATORY ADOPTION TIMELINE FOR POST-QUANTUM
CRYPTOGRAPHY (2026–2035)

Region	Policy	Status	Deadline	Scope
United States	NSA CNSA 2.0	Mandatory	2025/2030	National Security
United States	NIST IR 8547	Final	2026–2035	Federal agencies
European Union	ENISA/NIS2	Pending	2027	Critical infrastructure
Global	IETF RFC 9496	Voluntary	Adopted	Inter-operability
China	AOSQ	Draft	2028	Government systems

and supply chain participants. NIST IR 8547 provides migration guidance requiring risk assessments and transition plans by 2026, with full implementation scheduled through 2035 for non-national-security systems. The European Union’s NIS2 Directive mandates quantum-safe cryptography for critical infrastructure by 2027, though specific algorithm selections remain under ENISA evaluation [18]. Global IETF standards (RFC 9496) provide voluntary interoperability foundations that enable cross-border deployment but lack regulatory enforcement.

Organizations with international operations must navigate this fragmented landscape, potentially requiring jurisdiction-specific algorithm configurations and compliance documentation. The divergence between U.S. (CNSA 2.0 mandates specific algorithms) and EU (technology-neutral, risk-based approach) regulatory philosophies creates particular complexity for multinational ZTA deployments.

A. Phased Compliance Roadmap (2025–2030)

Based on the decision framework and regulatory timelines, we propose a phased implementation roadmap:

Phase 1 (2025–2026): Foundation. Deploy hybrid TLS (RFC 9496) across all external-facing services. Inventory classical cryptography assets. Pilot decision framework in non-critical internal services. Establish cryptographic agility infrastructure (algorithm negotiation, automated certificate rotation).

Phase 2 (2027–2028): Expansion. Implement ML-KEM for internal key establishment per framework risk tiering. Deploy ML-DSA for new certificate issuance. Procure hardware acceleration (AVX-512, FPGA) for high-throughput scenarios. Update IAM platforms for PQC support.

Phase 3 (2029–2030): Compliance. Complete CNSA 2.0 compliance for National Security Systems. Retire classical algorithms for key establishment. Fully operationalize framework-automated algorithm selection. Establish quantum-safe trust scoring for continuous risk evaluation.

IX. RESEARCH GAPS AND FUTURE DIRECTIONS

Despite rapid standardization progress, significant research gaps hinder optimal PQC-ZTA integration:

A. Critical Technical Gaps

- **Absence of Unified Integration Frameworks:** No standardized methodology exists for incorporating PQC into the complete ZTA lifecycle—from initial device provisioning through continuous authentication to credential revocation. Current deployments rely on ad hoc integration of point solutions. Our decision framework addresses this gap but requires toolchain implementation.
- **Limited Production-Scale Evaluation:** PQC algorithms have undergone extensive laboratory benchmarking but minimal evaluation in large-scale, heterogeneous ZTA networks with realistic traffic patterns, failure modes, and attack scenarios. Available data derives primarily from Google and Cloudflare deployments, which may not generalize to enterprise environments.
- **Insufficient Security Tooling:** Major IAM platforms (Okta, Azure AD, Ping Identity) and SIEM systems lack native PQC support, requiring expensive custom integration or vendor-dependent roadmaps that create lock-in risks.

B. Future Research Priorities

- 1) **Lightweight PQC for Real-Time ZTA:** Development of optimized parameter sets and implementations for resource-constrained endpoints, pending Falcon (FN-DSA) standardization for bandwidth-limited scenarios.
- 2) **Cryptographic Agility Toolchains:** Modular APIs and dynamic selection protocols enabling runtime algorithm negotiation without service interruption, including automated fallback and performance-based adaptation. Integration of our decision framework into service mesh control planes (Istio, Linkerd).
- 3) **Hardware Security Module Extensions:** Standardized TPM 2.0 PQC algorithm support with secure key generation, attestation, and sealing capabilities for edge device authentication.
- 4) **Quantum-Safe Trust Scoring:** Algorithmic frameworks for continuous risk evaluation that incorporate PQC algorithm strength, key freshness, and quantum threat timeline estimates into ZTA access decisions—extending our decision framework with dynamic risk assessment.
- 5) **Standardized Benchmarking Frameworks:** Industry-standard test suites for PQC performance in ZTA-specific scenarios (microsegmented mesh networks, high-frequency authentication, certificate chain validation) enabling rigorous cross-vendor comparison.

C. Limitations of Current Study

This survey necessarily relies on available performance data from major cloud providers (Google, Cloudflare) and NIST reference implementations. Longitudinal studies of PQC deployment in diverse enterprise ZTA environments—manufacturing, healthcare, finance—remain unavailable. The rapidly evolving standardization landscape, including pending FIPS 206 (Falcon) and ongoing IETF PKIX standardization, may alter specific algorithmic recommendations. Readers

should verify current standardization status before implementation.

X. CONCLUSION

The integration of Post-Quantum Cryptography into Zero Trust Architecture has transitioned from theoretical preparation to operational imperative following NIST standardization and NSA CNSA 2.0 enforcement. This paper's three-layer taxonomy provides a systematic framework for addressing multifaceted challenges spanning computational overhead, architectural constraints, and protocol compatibility. Our central contribution—a **decision framework mapping ZTA scenarios to optimal PQC algorithms**—transforms deployment from ad hoc experimentation to systematic engineering.

Analysis demonstrates that ML-KEM-768 provides viable performance for TLS 1.3 handshakes in enterprise ZTA, while ML-DSA signatures impose significant latency penalties requiring hardware acceleration for high-throughput scenarios. The divergence between resource-rich data centers and constrained IoT endpoints necessitates tiered deployment strategies, with hash-based signatures (XMSS, SLH-DSA) serving niche roles in long-term trust anchors where algorithmic conservatism outweighs performance requirements.

The 2030 CNSA 2.0 deadline for key establishment creates urgency, but premature optimization risks vendor lock-in and interoperability failures. Organizations should prioritize hybrid deployments enabling gradual migration, invest in cryptographic agility infrastructure, and operationalize our decision framework for context-aware algorithm selection. Only through such comprehensive approaches can we ensure that ZTA's security guarantees persist into the quantum computing era.

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